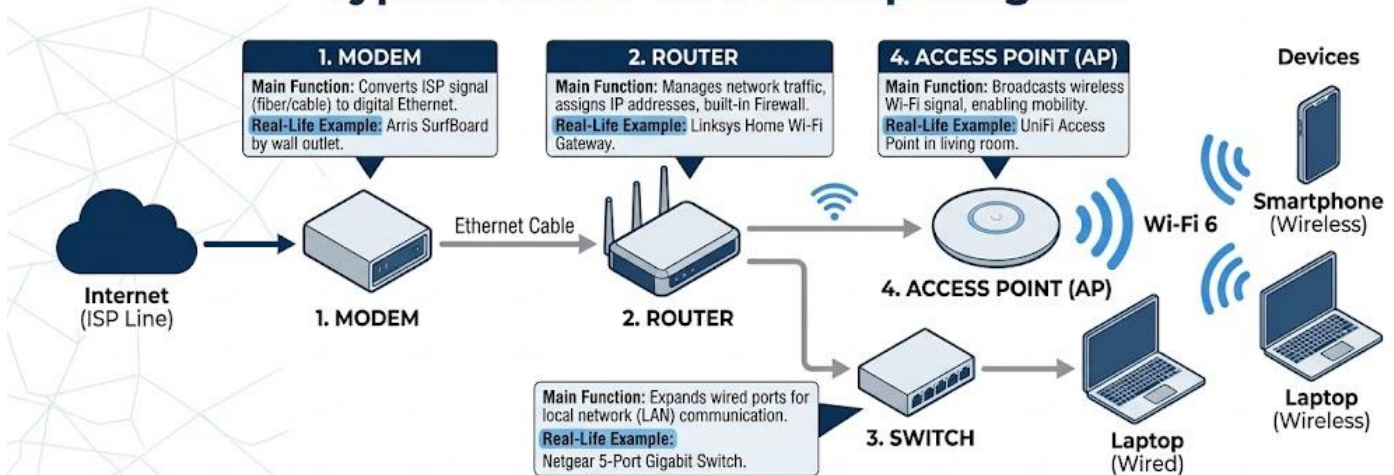


# Introduction to Networking Hardware Theory

## Task 1: Draw a Home Wi-Fi Setup Diagram

- **Goal:** Create a visual diagram showing how internet flows into a home.
- **What to include:**
  - **Internet Line (ISP)** → connects into the **Modem**.
  - **Modem** → connects to the **Router** with an Ethernet cable.
  - **Router** → connects to a **Switch** (for more wired ports).
  - **Switch** → connects to a **Wireless Access Point (AP)**.
  - **Devices:** Show an **Access Point** broadcasting Wi-Fi to a **smartphone**, and the **Switch** or **Router** connecting via cable or Wi-Fi to a **laptop**.

## Typical Home Wi-Fi Setup Diagram



Network Cable Types (Images & Usage)

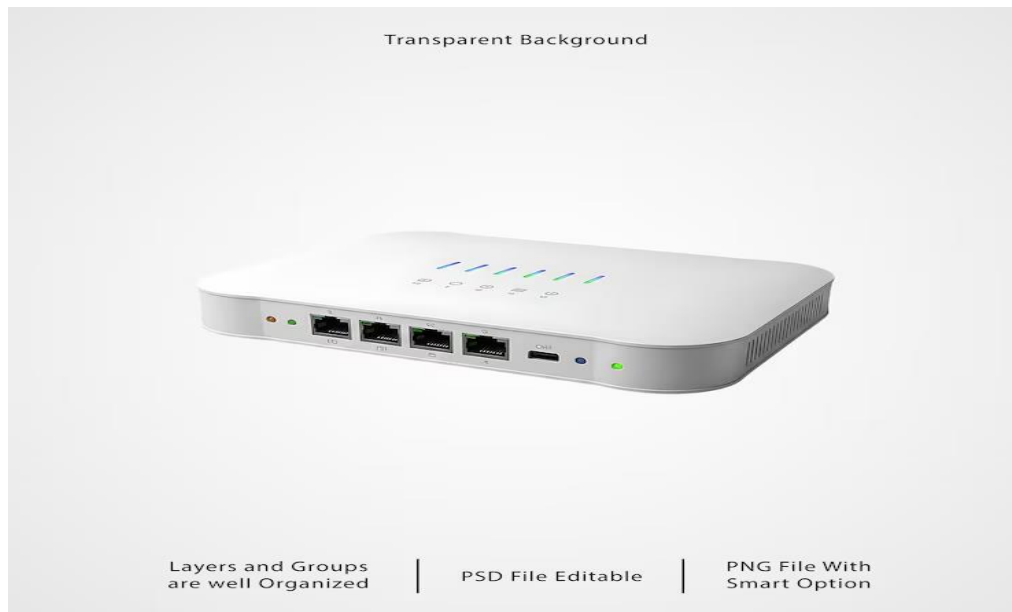
|  |                                                                                             |
|--|---------------------------------------------------------------------------------------------|
|  | <b>Ethernet Cable (Cat6)</b><br>Description: Short connections between Router, Switch, PCs. |
|  | <b>Coaxial Cable</b><br>Description: Primary link to older Cable Broadbands.                |
|  | <b>Fiber Optic Cable</b><br>Description: High-speed backbones and modern home broadband.    |

Router vs. Switch Comparison Table

| Feature          | Router                               | Switch                                 |
|------------------|--------------------------------------|----------------------------------------|
| Main Use         | Connects Home LAN to Internet WAN    | Connects devices inside the Home LAN   |
| Typical Location | Edge / Gateway of network            | Inside the LAN network rack/desktop    |
| Data Handling    | Routes packets via <b>IP Address</b> | Delivers frames via <b>MAC Address</b> |
| Direction        | Air Traffic Controller               | Local Delivery Driver                  |

## Task 2: Functions & Real-Life Examples of 5 Hardware Devices

- **Modem:**



- *Function:* Converts the incoming signal from your Internet provider (cable or fiber) into digital data.
- *Real-life example:* The small box by your wall outlet where the main Internet fiber or cable connects.

- **Router:**



- *Function:* Manages traffic, assigns local IP addresses, and directs data between your home network and the outside Internet.
- *Real-life example:* The main home Wi-Fi gateway device in your living room.
- 

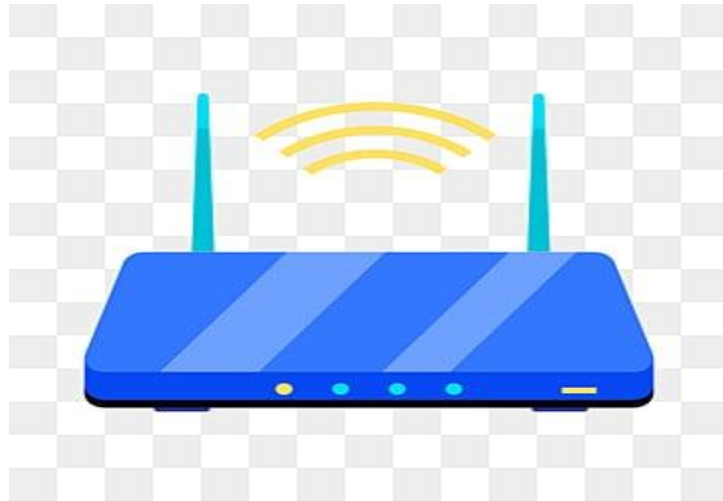
- **Switch:**



- *Function:* Expands wired network ports so multiple wired devices can communicate on the same local network without collisions.

- *Real-life example:* An office or computer lab network rack connecting 20 desktop PCs together.

- **Access Point (AP):**



- *Function:* Broadcasts a wireless Wi-Fi signal from a wired connection so wireless devices can join the network.
- *Real-life example:* White discs mounted on the ceiling in schools, hotels, or large offices to provide seamless Wi-Fi.

- **Network Cable:**



- *Function:* The physical medium (copper wires or glass fiber) carrying electronic or light signals between devices.

- *Real-life example:* The blue RJ45 Ethernet patch cable plugged into your PC or gaming console.

### Task 3: Three Network Cable Types (Images & Usage)

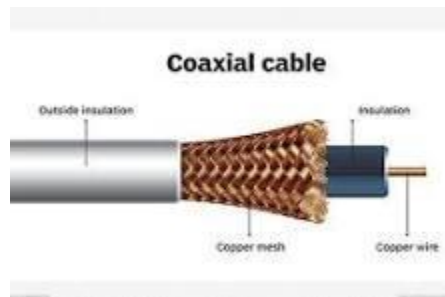
- **Ethernet Cable**  
**Pair):**



(Cat5e/Cat6 Twisted

- *Common use:* Connecting PCs, gaming consoles, smart TVs, and routers over short distances inside a home or office.

- **Coaxial Cable:**



- *Common use:* Connecting cable TV set-top boxes and older cable broadband modems to the wall jack.

- **Fiber Optic Cable:**



- *Common use:* High-speed backbone connections between cities, ISPs, and modern FTTH (Fiber-to-the-Home) broadband lines.

#### Task 4: Router vs. Switch Comparison Table

| Feature             | Router                                                                            | Switch                                                                                  |
|---------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Main Use            | Connects completely different networks (e.g., your Home LAN to the Internet WAN). | Connects multiple devices together within the same local network (LAN).                 |
| Typical Location    | At the edge or gateway of the network (where ISP enters).                         | Inside the local network (behind the router, in an office rack, or desk).               |
| How It Handles Data | Reads <b>IP addresses</b> (Layer 3) to route packets between networks.            | Reads <b>MAC addresses</b> (Layer 2) to deliver data frames directly to specific ports. |
| Traffic Direction   | Acts like an air traffic controller directing flights between different cities.   |                                                                                         |